Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/02/2007
PI Name:	Yu, Hongwei David Ph.D.		
Project Title:	Control of Bacterial Biofilm Formation in Space Exploration		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	25755-9320	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 6/30/2008, per A. Chu (9/07) NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC74G		
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Task Description:	Bacteria in nature and disease often choose a mode of growth known as a biofilm, which consists of numerous aggregating microorganisms (microcolonies) attached to a surface. Biofilm formation is a leading cause of bacterial pneumonia which can be acquired by astronauts during a long-term space mission and by patients with compromised immune systems such as those with cystic fibrosis. A capsular material called mucoid exopolysaccharide, or alginate is important in the formation of biofilms because it provides a key structural component for in vivo biofilms. The objective of this proposal is to study the genetic regulation of alginate in biofilm formation. The model organism is Pseudomonas aeruginosa, a ubiquitous biofilm-forming bacterium capable of displaying two distinctive phenotypes: a nonmucoid or mucoid colony morphology. The mucoid form, which is due to the overproduction of alginate, is a clinical indicator for a poor prognosis. While the two morphologies can be made inter-changeable under the laboratory conditions, little is known about the molecular mechanisms governing this process. Using genetics and functional genomics approaches, we will search for the biofilm “switches” on the genomes. The identified genes may become targets for the development of anti-biofilm therapeutics. The specific aims of this project are i) to identify and characterize the genes in the alginate regulation, ii) to elucidate the underlying mechanisms, and iii) determine what/how environmental signals modulate the expression of these regulators. Our long-term goal is to target bacterial own genes for the control of the space biofilm formation. The objective of the proposed research is to identify and characterize the biofilm-controlling genetic elements from the genomes of P. aeruginosa. It is our expectation that the approaches used here will lead us to identify the genes that control the formation of the mucoid biofilms. We will also provide new information regarding how these genes control the phenotypic switch, and what/how environmental signals may stimulate or inhibit the expression of these genes. These results will be significant, because they will provide genetic targets for anti-biofilm therapeutics, and aid in the future development of the gene-based space medicine. In addition, it is expected that the results will advance the emerging field of microbial biofilms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major Earth benefits of this proposal are to provide the basis for the future development of the new genetically defined drug with improved specificity and function and very little side effect.
Task Progress:	Using a global transposon screen, we have recently identified two genes of previously unknown function (PA4033, now designated mucE; and PA3649, now designated mucP) along with the known algW gene as positive regulators of alginate. We observed that overexpression of MucE caused an increase in alginate production and AlgU transcriptional activity and caused mucoid conversion of three different non-mucoid strains. Our studies of MucE determined that the protein is localized to the periplasm and that its C-terminus residues of WVF are required for its positive effect on AlgU activity and alginate production. These observations lead us to hypothesize that MucE was acting as a signal to activate a protease to cleave the anti-sigma factor MucA. These observations lead us to test whether the activation mediated by MucE required AlgW and MucP which were the closest homologs of E. coli DegS and YaeL/RseP in the P. aerugionsa genome. We observed that disruption of algW or mucP prevented the increase in alginate production and the algU activity caused by overexpression of MucE. Therefore we hypothesized that the C-terminus of MucE and perhaps other OMPs binds to the PDZ domain in AlgW and activates it to cleave MucA and that cleavage of MucA by AlgW would then make MucA a substrate for MucP which is predicted to be located in the inner membrane. However, MucE only caused an increase in alginate production and AlgU activity when it was highly expressed suggesting it may be the normal activator of AlgW. Genomic analysis revealed that two predicted OMPs (OprO and OprP) ended with a C-terminus of YVF which is similar to C-termini of MucE (WVF). Over-expression of MucE with YVF on its C-termini also caused an increase in alginate production and AlgU activity suggesting that the C-termini of OprO and OprP are likely to bind to the PDZ domain of AlgW and activate it to cleave MucA. Therefore we are proposing that the mechanism of regulation of AlgU is similar to ?E of E. coli, but that the signal that leads to the activation of AlgU is different from the signal that activates DegS to cleave RseA. DegS was activated best by proteins ending with YYF or YQF. We observed that over-expression of mutant forms of MucE ending in these residues either did not cause mucoidy conversion (YYF) or only a weak response (YQF) suggesting that Pseudomonas cells have evolved a response to proteins with somewhat different tails, but the significance of this observation is not known. Through this study, we have identified a protein signal (MucE) that can activate a signal transduction for mucoid biofilm formation. This result has been published in the May 7 2007 issue of PNAS. The citation of this paper is as follows: Qiu, D., V. M. Eisinger, D. W. Rowen, and H. D. Yu. 2007. Regulated proteolysis controls mucoid conversion in Pseudomonas aeruginosa. Proc. Natl. Acad. Sci. USA 104: 8107-8112. Submission of a patent application for the identification of a biofilm signal (MucE) in P. aeruginosa. Marshall University submitted a patent application for the discovery of MucE and the associated signal transduction pathway for alginate production. Dr. Niles and I have acquired an exclusive licence for this pending patent. We have created a start-up biotech company called Progenesis Technologies, LLC which plans to mass produce high-quality bacterial alginate. Currently, alginate is made from seaweeds. The large-scale sea farming of seaweeds poses a direct threat to the environment. In addition, the level of polymerization, monosaccharide polydispersity of seaweed alginates cannot be controlled. Bioprocessing and separation of certain types of seaweed alginates are expensive. The objective of Progenesis is to produce alginates through genetically defined bacterial strains. Progenesis is in the final phase of capitalization.
Bibliography Type:	Description: (Last Updated: 04/09/2020)
Articles in Peer-reviewed Journals	Wilson KR, Napper JM, Denvir J, Sollars VE, Yu HD. "Defect in early lung defence against Pseudomonas aeruginosa in DBA/2 mice is associated with acute inflammatory lung injury and reduced bactericidal activity in naive macrophages." Microbiology. 2007 Apr;153(Pt 4):968-79. PMID: 17379707 , Apr-2007
Articles in Peer-reviewed Journals	Qiu D, Eisinger VM, Rowen DW, Yu HD. "Regulated proteolysis controls mucoid conversion in Pseudomonas aeruginosa." Proc Natl Acad Sci U S A. 2007 May 8;104(19):8107-12. PMID: 17470813 , May-2007

Articles in Peer-reviewed Journals	Poschet JF, Timmins GS, Taylor-Cousar JL, Ornatowski W, Fazio J, Perkett E, Wilson KR, Yu HD, de Jonge HR, Deretic V. "Pharmacological modulation of cGMP levels by phosphodiesterase 5 inhibitors as a therapeutic strategy for treatment of respiratory pathology in cystic fibrosis. " Am J Physiol Lung Cell Mol Physiol. 2007 Sep;293(3):L712-9. Epub 2007 Jun 22. PMID: 17586695 , Sep-2007
Awards	Yu HD. "Marshall University Distinguished Artists & Scholars Award (Senior Recipient for Excellence in Sicence & Technology), April 2007." Apr-2007
Dissertations and Theses	Head NE, Yu HD. "Regulation of Biofilm Formation of Pseudomonas aeruginosa." Dissertation, Marshall University, Huntington, WV, May 2006. , May-2006
Journal/Magazine covers	Sollars VE, Yu HD. "Paper featured on the cover of Microbiology (Defect in early lung defence against Pseudomonas aeruginosa in DBA/2 mice is associated with acute inflammatory lung injury and reduced bactericidal activity in naive macrophages.)." Microbiology 2007 Apr; Volume 153, part 4. , Apr-2007
Papers from Meeting Proceedings	Yu HD. "Development of Mucoid Pseudomonas aeruginosa Biofilms in Cystic Fibrosis." Star Symposium 2007, Morgantown, WV, September 17-18, 2007. Pending, Star Symposium 2007, September 17-18, 2007. , Sep-2007
Patents	US Patent Appl. No. 60/787,497. Pending, March 2006. Mar-2006 Yu HD, Qiu D. "Methods of Detecting and Controlling Mucoid Pseudomonas Biofilm Production."
Significant Media Coverage	Aulick HL. "Researcher Profile: Hongwei Yu, Marshall University." The Neuron, West Virginia Journal of Science and Research, Summer 2007, p. 5., Jun-2007
Significant Media Coverage	Hardiman JT. "MU announces official launch of Progenesis." The Herald-Dispatch (newspaper), May 27, 2007., May-2007
Significant Media Coverage	Wellman D. "Marshall faculty member’s article published in PNAS." Marshall University Press Release May 02, 2007 and also reported in the Herald-Dispatch, May 2007., May-2007